

Backend Deployment Readiness Checklist

Item	Resego	Olwethu
Source code committed to GitHub	☐	
requirements.txt included	☐	
Backend entry point identified	☐	
Framework documented	☐	
Project runs from a clean clone	☐	
README updated with setup instructions	☐	
Azure deployment ready	☐	☐

Checklist Responses

1. Was the requirements.txt generated? Yes, it has been generated for the project to ensure all necessary libraries are tracked.

2. How did they generate requirements.txt? "We approached the creation of the requirements.txt manifest through a manual configuration and dependency mapping process. Given that the project requires a specific, controlled set of libraries to function

correctly, I manually curated the manifest to ensure the integrity of the backend environment.

Here is the step-by-step breakdown of your process:

- **Dependency Identification:** I conducted an audit of the current application structure and identified all the specific libraries and frameworks (such as Flask) that the backend relies on for its core functionality.
- **Version Mapping:** I cross-referenced the necessary versions required for our deployment environment (Azure) to ensure that the dependencies listed are compatible and stable.
- **Manual Manifest Construction:** I performed a manual creation of the `requirements.txt` file within VS Code, explicitly documenting each required component to prevent any "dependency hell" or version conflicts during the deployment phase.
- **Validation:** After creating the file, I verified that these dependencies align with our project's architecture, ensuring that the backend will initialize correctly when the application is launched.

3. Why is it needed? It is essential for maintaining environment consistency. It acts as a manifest that allows any developer or cloud server to install the exact same set of tools used during development, preventing compatibility errors when the application is moved to a new environment.

4. What does the application's entry point do? The entry point (typically `app.py`) serves as the core of the application. It initializes the Flask framework, configures settings such as database connections, and defines the URL routes that allow the website to respond to user requests.

5. How would someone else run the project from a fresh computer? To run the project on a new system, follow these steps:

- **Clone the repository:** Download the project code from GitHub.
- **Set up the environment:** Create a new virtual environment using `python3 -m venv venv` to isolate the project.

- **Install dependencies:** Run `pip install -r requirements.txt` to automatically install all required libraries.
- **Run the application:** Execute `python3 app.py` to start the backend server.



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